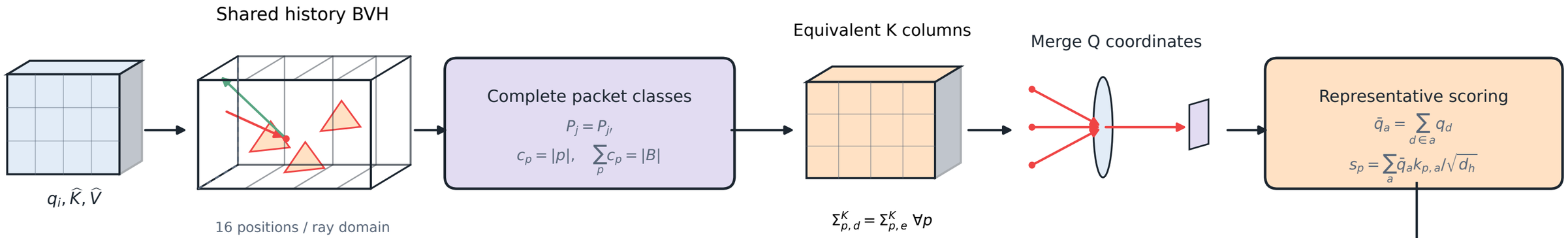
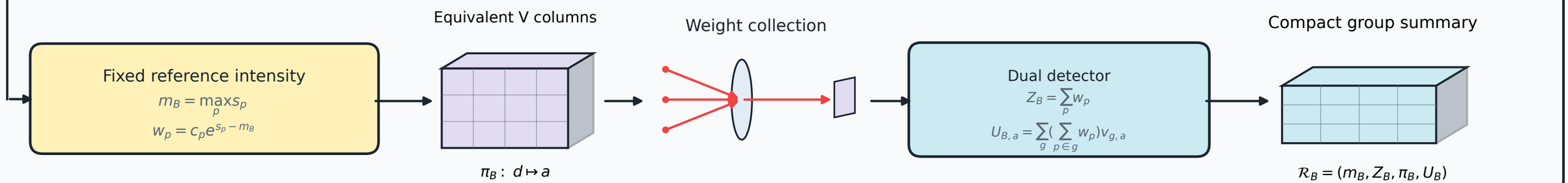
Ray Tracing Attention

Shared geometry, representative coordinates and complete softmax summaries

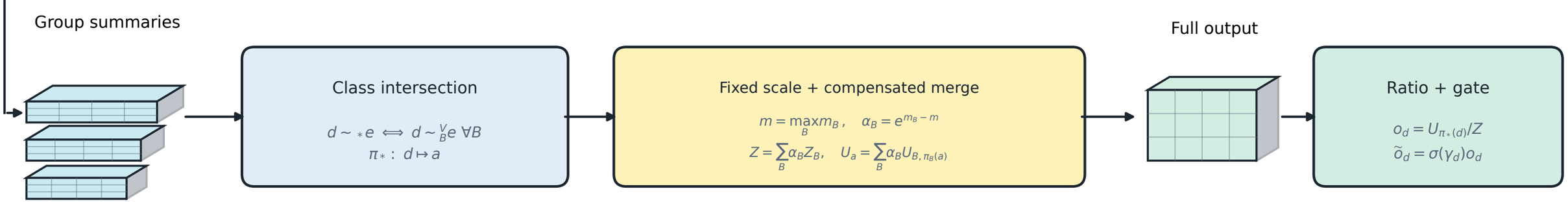
(a) Shared causal intersections and exact coordinate classes



(b) Group intensity and representative content



(c) Intersect group classes, merge once, restore every output coordinate



$p = \mathrm{fl}(ab), \quad r = \mathrm{fma}(a, b, -p)$ Compensated Q, weights, reference and content sums